

Registration Number: _____

Name: _____

Branch & Section: _____



SRM UNIVERSITY – AP, ANDHRA PRADESH

End Term Examination, Dec 2025

[Question Paper ID: 010869]

Subject : ENGINEERING PHYSICS

Title

Batch : 2025

Degree : B.Tech.

Branch : CSE, CSE AIML, ECE, EEE, ME&SC

Subject : FIC 102

Code

Max Marks : 50

Duration : 2 hours

QP Set : 2

Instructions:

Scientific calculators are allowed. Use the value of acceleration due to gravity, $g = 10 \text{ m/s}^2$, wherever required.

PART-A ($5 \times 3 = 15$ Marks)

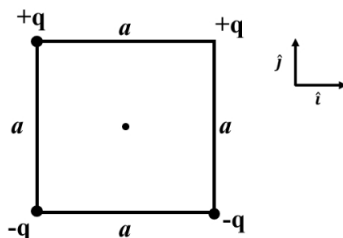
Answer **all** Questions

Marks CO BL

- | | | | | |
|----|--|---|-----|---|
| 1. | A bomb of mass 16 kg at rest explodes into two pieces of masses 4 kg and 12 kg . The velocity of 12 kg mass is 4 m/s . Determine the kinetic energy of the 4 kg mass. [3] | 3 | 2 | 2 |
| 2. | The vector $\vec{A}$ is represented by $a\hat{i} + \sqrt{3}a\hat{j}$. [2+1]
a) If the magnitude of the vector is 20, Find the value of a .
b) Find a unit vector along $\vec{A}$. | 3 | 2 | 2 |
| 3. | a) Calculate the energy of a photon of light having a wavelength of 400 nm .
b) If this light falls on a zinc surface (work function = 3.6 eV), determine the stopping potential (in volts) required to stop the fastest photoelectrons emitted. [1+2] | 3 | 3 | 3 |
| 4. | A traveling wave propagates according to the expression $y(x, t) = 8 \sin(40t - 0.1x)$, where y and x are in meters and t in seconds.
a) Determine the <u>wavelength</u> . [1+2]
b) What will be the <u>speed of the wave</u> ? | 3 | 2,3 | 3 |

5. Four charges $+q$, $+q$, $-q$ and $-q$ are situated at the corners of a square (of side a), as shown in figure. 3 3 3
[2+1]

- a) Find the electric field vector and its magnitude at the center of the square.
b) Also, determine the electric potential at the center.



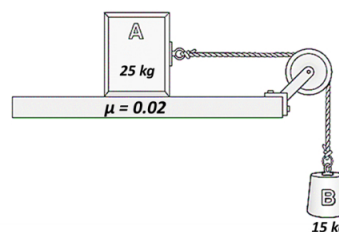
PART-B ($5 \times 7 = 35$ Marks)

Answer any 5 Questions

Marks CO BL

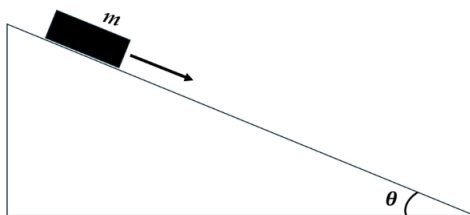
6. Two blocks of masses $m_A = 25$ kg and $m_B = 15$ kg are tied to the ends of a string which is massless and passes through a frictionless pulley as shown in figure. The coefficient of friction between the block A and the table is 0.02 7 2 2
[2+3+2]

- a) Draw the free body force diagram for the blocks.
b) Using Newton's laws, find the acceleration of the blocks.
c) What will be the tension in the string?



7. A block of mass $m = 4\sqrt{2}$ kg is sliding along a slope ($\theta = 45^\circ$) of a rough surface ($\mu_k = 0.2$) as shown in the figure. The block moves a distance of $d = 1$ m. 7 3 3
[5+2]

- a) Find the work done by normal force, frictional force, and gravity, while the object is sliding down the inclined plane.
b) Calculate the total work done by the three forces.

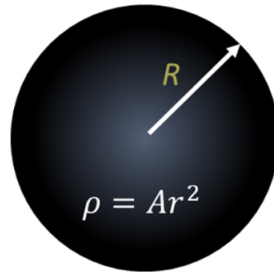


8. The wavelength of the light source used in the double slit experiment is 530 nm. The distance between the viewing screen and the double slit is 1.2 m. The distance between the two slits is 0.030 mm. 7 2 2
[3+2+2]

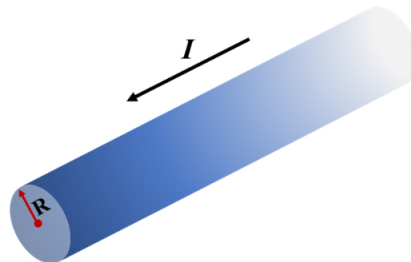
- a) Calculate the position of the second-order bright fringe ($m = 2$) from the center line.
b) Calculate the distance between adjacent bright fringes.
c) Calculate the distance between adjacent bright and dark fringes.

9. (a) What do you mean by the work function of a metal? [2+2+3] 7 2,3 3
 (b) Calculate the critical (or threshold) wavelength and critical frequency for photoelectron emission from a copper metal surface. The work function of copper is 4.65 eV.
 (c) Calculate the maximum kinetic energy of the electrons and the stopping potential when light of wavelength 220 nm falls on copper surface.

10. a) What is Gauss Law of electrostatics? [2+5] 7 2 2
 b) A non-conducting Solid sphere carries a non-uniform volume charge density $\rho = Ar^2$, where A is a constant and r represents the distance from the center of the sphere. Find its electric field $\vec{E}$ at any point inside ($r < R$) and outside ($r > R$)



11. There is an infinite conducting wire of radius R carrying current I as shown in the figure. [5+2] 7 3,4 3
 a) Find out the magnetic field $\vec{B}$ inside ($r < R$) and outside ($r > R$) the wire .
 b) Plot the magnitude magnetic field (B) vs distance (r).



12. a) What is the use of capacitor in an electrical circuit? [2+3+2] 7 3 3
 b) Derive the *expression of capacitance* for a parallel plate capacitor with area of each plate as A and the distance between the plates as d .
 c) Calculate the capacitance for a parallel-plate capacitor in vacuum with $A = 4\text{m}^2$ and $d = 4\text{mm}$.

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